

Polynomials – 6

Polynomials over finite fields

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Theoretical minimum.

Theorem 1 (counting roots). A non-zero polynomial of degree n over a field has at most n roots. Over $\mathbb{F}_q$ one has $x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$, the multiplicative group $\mathbb{F}_q^*$ is cyclic, and a function $\mathbb{F}_q \rightarrow \mathbb{F}_q$ is given by a unique polynomial of degree less than q .

Theorem 2 (Frobenius and irreducibles). The map $\varphi(x) = x^p$ is a field automorphism of $\mathbb{F}_q$ with $q = p^n$, fixing exactly $\mathbb{F}_p$. An irreducible polynomial of degree d over $\mathbb{F}_p$ divides $x^{p^n} - x$ if and only if $d \mid n$; hence the number of monic irreducibles of degree n is $\frac{1}{n} \sum_{d \mid n} \mu(d) p^{n/d}$.

Fact 1 (derivative in characteristic p). Over $\mathbb{F}_p$ the derivative of x^p is zero, so $f' = 0$ means f is a polynomial in x^p , and then f is a p -th power. This is why the criterion “ f is squarefree iff $\gcd(f, f') = 1$ ” behaves differently from the characteristic zero case.

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1. Prove Theorem 1 in full, prove that $\mathbb{F}_q^*$ is cyclic, and deduce Wilson’s theorem and the criterion for a to be a k -th power in $\mathbb{F}_q$.
 2. (RMO) Let K be a finite field. Find all integers $n \geq 2$ with the property that every polynomial of degree n in $K[X]$ having no root in K is irreducible over K .
 3. (RMO 2004) Let $f \in \mathbb{Z}[X]$ and for $n > 2$ define $f_n: \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ by $f_n(\bar{x}) = \overline{f(x)}$.
 - a) Prove that f_n is well defined.
 - b) Find all $f \in \mathbb{Z}[X]$ for which f_n is surjective for every $n > 2$.

4. (IMC) Let p be prime. Call a positive integer n interesting if $x^n - 1 = (x^p - x + 1)f(x) + pg(x)$ for some $f, g \in \mathbb{Z}[x]$.
- a) Prove that $p^p - 1$ is interesting.
 - b) For which p is $p^p - 1$ the smallest interesting number?
5. (Putnam 2002) Let p be a prime. Prove that the determinant of the matrix with rows (x, y, z) , (x^p, y^p, z^p) and $(x^{p^2}, y^{p^2}, z^{p^2})$ is congruent modulo p to a product of polynomials of the form $ax + by + cz$ with integer a, b, c .